Motion-borne geometry: when an information metric needs the dynamics beneath it

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A stable emergent structure can relate to its substrate in two ways: it can be merely *instantiated* — some motionless configuration of the substrate would carry the same structure — or it can be *actively maintained*, existing only because the substrate moves. We introduce an operational test that separates the two, and apply it in a preregistered numerical study of the mutual-information-graph metric of finite spin chains, across fixed-point, quasiperiodic, chaotic, metastable, localized, and driven (discrete-time-crystal) dynamics. The test has three probes: witnessed motion beneath a stationary metric, response to switching the dynamics off, and an explicit search for a motionless state carrying the same time-averaged structure. The outcome is a single-survivor split. Every class's time-averaged metric can be reproduced by some stationary state — chaotic classes almost exactly, consistent with eigenstate thermalization — except the quasiperiodic class: across 80 runs at two system sizes, no stationary state within a broad smooth-of-H family comes within the stationarity threshold. The same class is independently singled out by its response to weak dephasing, which *pins* its moving metric to the time average — a negative drift ratio stable over two decades of noise strength — while destabilizing chaotic-class metrics. Incommensurate coherence-carried motion is thus the one regime, in this family, whose emergent structure demonstrably needs its dynamics. The study was preregistered with committed seed manifests and a dated amendment log; its discipline caught four failure modes before publication, including the discovery that the out-of-time-order correlator — the field's default dynamics diagnostic — fails a basic witness requirement: it fires on frozen states. We report the corrected instruments, the corrections themselves, and class-resolved scope walls (including a measured, class-dependent partition-dependence of the metric).

1 Introduction

A stable emergent structure can stand in two different relations to the substrate that carries it. It can be *merely instantiated*: some motionless configuration of the substrate would carry the same structure, and the observed dynamics is incidental to it. Or it can be *actively maintained*: the structure exists only because the substrate moves, and no frozen configuration reproduces it. The distinction is easy to state and surprisingly hard to test — most diagnostics of "underlying dynamics" do not actually certify that anything is moving, and most claims of "the dynamics sustains the structure" are never confronted with a serious search for a motionless impostor.

This paper builds that test and runs it, in the most controlled setting we could construct: finite spin chains, where the emergent structure is the mutual-information-graph metric of Refs. [1, 2, 3] — pairwise mutual information converted to link weights and shortest-path distances on a posited site factorization — and the substrate dynamics ranges over fixed-point, quasiperiodic, chaotic, metastable, localized, and driven (discrete-time-crystal [4, 5]) classes. Prior work on such information-metric structures has been static: ground-state network robustness under projective attacks [2], metricity as a phase diagnostic [3]. Our object is the *time dependence* of the structure over a moving state, and its need

— or lack of need — for that motion.

The test has three probes. A *witness battery* certifies that the state moves while the metric does not, under a requirement we treat as non-negotiable: a witness of motion must vanish identically on a frozen state. A *switch-off response* measures what happens to the metric when the motion is killed in place. And a *motionless-comparator search* optimizes explicitly over stationary states for one that carries the same time-averaged metric. The three probes are preregistered — every metric, threshold, and analysis rule fixed before code was written, every run seeded from committed manifests, every post-hoc change a dated amendment (Appendix C) — because a study of this shape has one dominant failure mode, self-deception through flexible metric choice, and we preferred to make our mistakes in public.

The mistakes were made, caught, and are reported as results. Four times, the discipline fired: a registered class certificate was mis-specified (Poisson statistics for a free chain); the out-of-time-order correlator — the field's default dynamics diagnostic — failed the null test, firing on a frozen ground state, and was discarded by its own preregistered clause; an early robustness comparison was exposed by our adversarial reanalysis as a floored-denominator artifact; and a narrow first version of the comparator search over-claimed unmatchability for two classes that a hardened search then matched. What survives this gauntlet is correspondingly compact:

1. **An operational test** of instantiated-vs-maintained emergent structure (witnesses with null-silence, switch-off, comparator search), portable to any system with a computable structural functional.

2. **A single-survivor measurement.** Every class's time-averaged metric is carried by some stationary state — chaotic classes almost exactly, consistent with eigenstate thermalization — except the quasiperiodic class: 0 of 80 runs matchable at two system sizes, robustly across mode numbers and optimizer strength.

3. **A sign structure.** Weak dephasing *stabilizes* the moving metrics of coherence-carried classes (negative log drift ratio, flat over two decades of noise strength) while destabilizing chaotic-class metrics — the same class singled out by the comparator search is singled out by noise. The mechanism is standard decoherence damping [6]; the class-resolved geometric reading appears, within the scope of our survey [2, 3, 7, 8], to be unreported.

4. **A methods record.** The null-silence filter and the fresh-seed, power-stabilized adjudication of sharp statistical criteria — with the OTOC's failure and the seed-lottery episode as measured exhibits, and the complete dated amendment log as Appendix C.

5. **A driven-phase case study** in which the sustained-by question resolves *negatively* despite maximal appearances: the discrete-time-crystal metric survives removal of its drive unchanged — localization, not the subharmonic motion, holds it.

What this paper does not claim. The metric is an information-theoretic construct on a posited factorization; its factorization dependence is measured (Sec. 8) and it is not spacetime geometry — no claim here touches gravity or holography. No mechanism is claimed as new physics. No assumption is made that information is ontologically primitive. All dynamics is stated against an external laboratory clock. We make no priority claims about preregistration in computational physics; the artifact is offered, not its firstness. The model family is small ($N \leq 14$) and the conclusions are in-model.

Section 2 defines the family, functional, witnesses, and protocol; Secs. 3–7 report results; Sec. 8 states the scope walls; Sec. 9 discusses what the test opens up.

2 Models, functional, witnesses, and protocol

2.1 Model family

All models are open chains of N qubits with $\hbar = 1$ and the nearest-neighbour coupling setting the unit of energy and time. Every dynamical statement below is made with respect to an external laboratory clock; the drive of the Floquet family defines a second, stroboscopic clock derived from it. Nothing in this study bears on the question of clocks internal to a system, which we regard as out of scope.

We use three tracks. Track A collects four closed-system classes chosen to span the qualitative types of quantum motion (Table 1).

Track C reuses the chaotic and quasiperiodic Hamiltonians as "scrambling" and "integrable" regimes with Néel-anchored product-state ensembles, and adds a localized regime (XXZ with random fields, $\Delta = 1$, $W = 8$; 20–40 disorder realizations; $\langle r \rangle = 0.38$–$0.40 \in [0.36, 0.42]$). Track B is a kicked-Ising discrete time crystal modeled on the established protocol [4, 5]: an imperfect global π-flip (ε the imperfection) alternating with disordered Ising interactions, 20 disorder realizations $\times$ 5 initial product states, with two comparator regimes — no interactions (r1) and no disorder (r2).

The class certificates are *preregistered sanity checks*. One failed as originally registered: we specified a Poisson level-statistics window for the integrable class, which is a heuristic for generic integrable models and is inapplicable to a free chain whose many-body spectrum is exactly the subset-sums of its single-particle spectrum. The replacement certificate — reconstruction of the many-body spectrum from single-particle energies to 10^{-8} — is strictly stronger, and the misspecification plus its dated repair are part of the study record (Appendix C). A second certificate finding: the quasiperiodic incommensurability requirement is *exhaustively unsatisfiable* at $N = 8$ (0 of 56 magnon triples pass), which fixed the usable size range $(10, 12)$ for class comparisons.

2.2 The information metric and its stationarity

From the state (pure or mixed) we compute all one- and two-site reduced density matrices, the pairwise mutual information $I_{ij} = S_i + S_j - S_{ij}$, and normalize $x_{ij} = I_{ij}/(2 \ln 2)$. Following the emergent-metric construction of Ref. [1], link weights are $w_{ij} = -\ln x_{ij}$, regularized by a floor $x_{\min} = 10^{-6}$, and the *metric* $D(t)$ is the matrix of weighted shortest-path distances on the complete graph over sites (Fig. 1). We emphasize the scope of the word "metric" here: D is the metric structure of the mutual-information graph on a *posited* site factorization. It is not a claim about spacetime, and its dependence on the factorization is measured, not assumed away (Sec. 8).

Stationarity is judged on the window $\mathcal{W} = [20, 200]$ (units of inverse coupling; stroboscopic periods for Track B), sampled every $\Delta t = 0.5$: with $\bar{D}$ the window mean and $\delta\Phi(t) = \|D(t) - \bar{D}\|_F / \|\bar{D}\|_F$, the metric is stationary iff $\max_{t \in \mathcal{W}} \delta\Phi(t) < \varepsilon_\Phi = 0.25$. The threshold was set by a preregistered calibration pilot that measured the construction's finite-size fluctuation floor (the originally registered 0.05 sat below the noise floor of every dynamical class — an instrument artifact the pilot existed to catch; Appendix C, Amendment 3). A cap diagnostic — the same drift restricted to above-floor links — is reported alongside every verdict to expose any cap dominance; none of the verdicts below is cap-dominated.

2.3 Witnesses and the null-silence requirement

The claim "the metric is stationary while the state moves" is only as good as the certificate of motion. We require every *witness of motion* to satisfy a null-silence condition: **it must vanish identically on a frozen state.** The null comparator is class (i) — an eigenstate, whose evolution is a global phase — and the preregistered rule is that any witness which fires on the null is discarded as a witness, whatever its other virtues.

The battery: (W1) the participation ratio PR_A of the binned Bohr spectral measure

$$A(\omega) \;=\; \sum_{m,n} |c_m|^2 \, |c_n|^2 \; \delta(\omega - |E_m - E_n|), \quad (1)$$

equal to 1 exactly on the null; (W2) the recurrence distance $d(t) = 1 - |\langle \psi(t_0) | \psi(t) \rangle|^2$, identically zero on the null, summarized by its window mean; (W4) the off-diagonal energy coherence $\Xi = \sum_{E_m \neq E_n} |\rho_{mn}|^2$, computed over energy-*distinct* pairs so that coherence within a degenerate level — which does not move — does not fire it; and, for Track B, (W5) the subharmonic Fourier weight of the stroboscopic magnetization at half the drive frequency, with its rigidity curve $h_{\mathrm{sub}}(\varepsilon)$. All are invariant under global phase and basis relabelings in the senses catalogued in Appendix D.

We also registered (W3) an out-of-time-order correlator, $C(r,t) \;=\; \frac{1}{2}\langle |[\sigma^z_{i_0+r}(t), \sigma^z_{i_0}]|^2 \rangle$, with saturation value and arrival time as statistics. It failed the null test — measured, not argued: on the frozen fixed point it rises to $C \approx 1.11$

class	Hamiltonian	initial ensemble	class certificate
(i) fixed point	transverse-field Ising, $H = -\sum_i \sigma_i^z\sigma_{i+1}^z - g\sum_i \sigma_i^x$, $g = 1.5$	ground state (single deterministic run)	gapped, nondegenerate
(ii) quasiperiodic	XX chain, $H = \frac{1}{2}\sum_i(\sigma_i^x\sigma_{i+1}^x + \sigma_i^y\sigma_{i+1}^y)$	20 superpositions of 3 single-magnon modes, pairwise-incommensurate Bohr gaps	free-spectrum reconstruction $< 10^{-8}$
(iii) chaotic	mixed-field Ising, $(h_x, h_z) = (0.9045, 0.8090)$	20 Haar-random product states	$\langle r\rangle = 0.529$–$0.542 \in [0.51, 0.55]$
(iv) metastable	ferromagnetic Ising, $g = 0.05$, dressings $\delta g_i \in [-0.01, 0.01]$	$\lvert\uparrow\cdots\uparrow\rangle$, 20 dressings	quasi-degenerate doublet

Table 1: Track-A classes. Certificates are preregistered sanity checks, run at every system size used before any confirmatory analysis.

by $t = 1.5$. The OTOC certifies Heisenberg-*operator* spreading, which a gapped ground state supports; it does not certify state motion. Under the preregistered rule its statistics were discarded from the class-separation criterion, with consequences reported in Sec. 4. We keep it in the outputs as the operator-spreading diagnostic it actually is.

2.4 Comparators and perturbation protocols

Three controls give the sustained-by test its content. The *diagonal ensemble* $\bar\rho = \sum_n |c_n|^2 |n\rangle\langle n|$ is the canonical motionless partner of a run: exactly stationary, with the same time-averaged two-site reduced density matrices. We registered it as "matched to the time-averaged metric by construction" — which is false, and measurably so: mutual information is nonlinear in the (correctly time-averaged) reduced density matrices, and $\Phi[\bar\rho]$ sits 43–90% away from $\bar D$ depending on class. The corrected comparator methodology — an explicit optimization over stationary states — is described with its results in Sec. 6. The *switch-off test* dephases the state in the energy eigenbasis mid-window (equivalently: projects onto $\bar\rho$, since populations are conserved) and asks whether the metric's behaviour changes. And three *perturbation protocols* — a local Hamiltonian quench $\lambda\sigma^z$, a local dephasing channel of rate γ, and the loss of two non-adjacent sites — probe robustness through the scale-free log drift ratio

$$\log\rho \;=\; \log\left[\frac{\max_{t>t_p}\delta\Phi_{\rm pert}(t)}{\max\big(\max_{t>t_p}\delta\Phi_{\rm unpert}(t),\,10^{-3}\big)}\right], \tag{2}$$

with strengths $(\lambda, \gamma) = (0.1, 0.01)$ fixed by the calibration pilot before any confirmatory run.

2.5 Preregistration protocol and statistics

Every metric, threshold, witness, comparator, and analysis rule above was fixed in a specification document before implementation code was written; every run consumed seeds from manifests committed to the repository before execution; and every post-hoc change is a dated entry in an amendment log, reproduced in full as Appendix C with commit hashes. The study ran as: specification → threshold review → calibration pilot (exploratory, excluded from confirmatory statistics) → confirmatory campaign → an adversarial companion analysis tasked with attacking our own conclusions → a witness-battery reformalization forced by the null-test failure, itself confirmed on fresh seeds. Class separation is judged by exact Mann–Whitney AUC ≥ 0.95 for at least two witness statistics per class pair at two system sizes (singleton classes by exact-value exclusion); robustness differences by $|\Delta\,\overline{\log\rho}| > \ln 1.5$ with disjoint BCa bootstrap confidence intervals, replicated with consistent direction at two sizes; disordered ensembles resample at the realization level. Where a sharp threshold met a finite ensemble, seed sensitivity was measured and stabilized ($n = 40$, Sec. 4); where it met a third size, the trend was checked descriptively ($N = 14$, Appendix B).

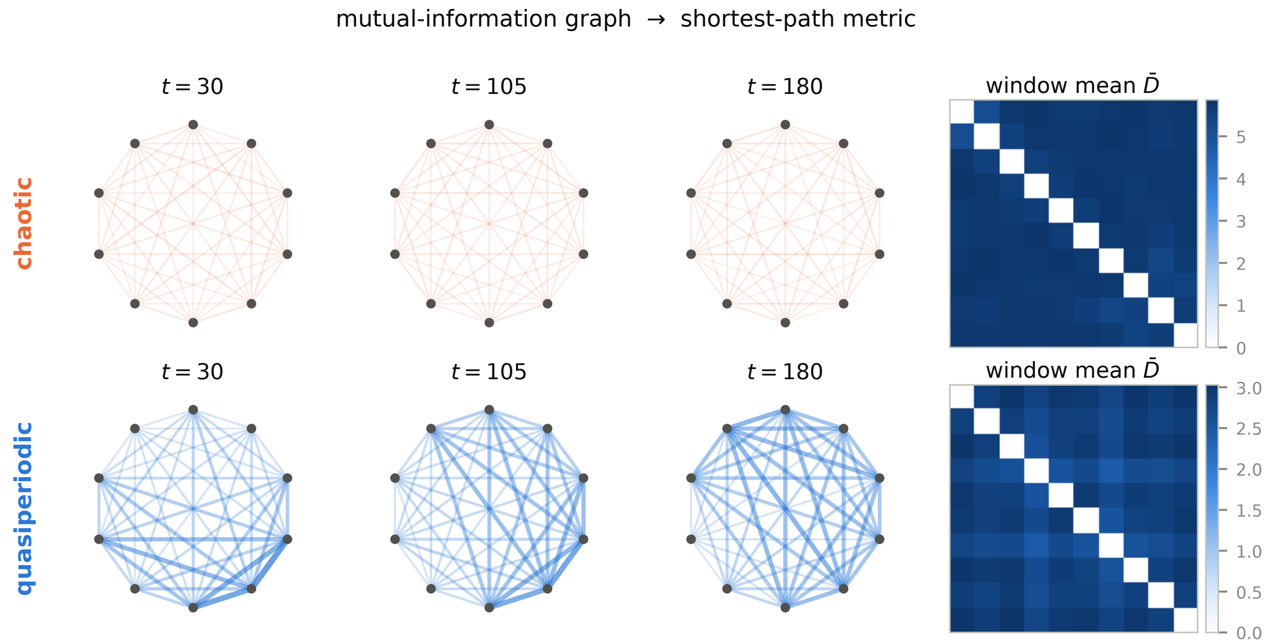

Figure 1: The pipeline, on real data (representative $N = 10$ runs). Nodes are sites; edge weight is pairwise mutual information; the metric D is the shortest-path distance matrix. Top: a chaotic-class state — the graph barely changes across the analysis window (its weak, near-uniform mutual information yields the large, static distances of the window mean $\bar{D}$). Bottom: a quasiperiodic state — the graph visibly rearranges as the incommensurate magnon beats evolve.

3 Results I: stationary metrics over witnessed motion

The baseline campaign establishes the phenomenon the rest of the paper interrogates: metrics that do not move over states that demonstrably do (Fig. 2).

At the calibrated threshold $\varepsilon_\Phi = 0.25$, the chaotic and scrambling ensembles are metric-stationary in every run (20/20 at each size; max window drift 0.09–0.21) while their motion witnesses read near maximum: $\Xi \geq 0.96$, recurrence mean $\approx 1 - 1/D$, and Bohr participation ratios in the hundreds to thousands. The driven track is the sharpest instance: at drive imperfection $\varepsilon = 0.03$, all 100 runs are metric-stationary while the subharmonic witness reads 0.939 — a locked period-doubled magnetization response over a frozen mutual-information metric. At the other extreme, the quasiperiodic and metastable classes have genuinely moving metrics (max drift 0.53–1.61; no run stationary), the integrable regime likewise (0.35–0.64), and the fixed point is stationary to machine precision (2×10^{-13}) with every witness silent — the null behaving as a null. The localized regime straddles the threshold (80/100 stationary at $N = 12$, drift 0.14–0.34) and is treated as a boundary case throughout.

Two honesty notes frame everything that follows. First, for the chaotic classes, metric stationarity over motion is *expected*: the metric is built from two-site observables, and the equilibration of local observables in quenched chaotic systems is standard physics. The existence table is the setup, not the result. Second, none of these verdicts is an artifact of the weight floor in the metric construction: the above-floor drift diagnostic tracks the full-graph drift in every class, and the mean graphs have no floor-dominated links.

4 Results II: the witness discipline at work

This section reports the study's methodological core: what the preregistered witness requirements did when the data pushed back. We present it as a dual record — the original registration failed; the ratified repair passed on fresh seeds — because both halves are results (Fig. 3).

The OTOC is not a witness of motion. The

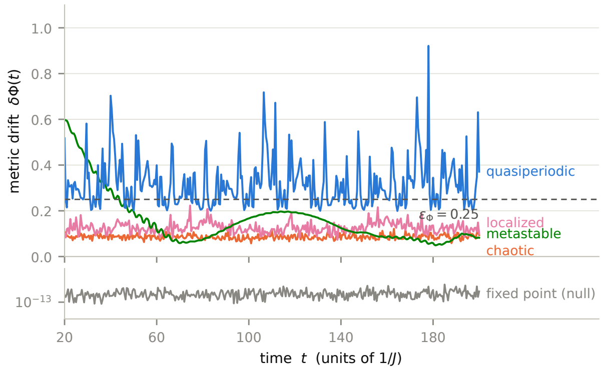

Figure 2: Metric drift $\delta\Phi(t)$ for representative runs ($N = 10$). The chaotic metric is stationary under the preregistered threshold while the quasiperiodic metric oscillates far above it; the metastable class shows its slow coherent wave; the localized run sits near the threshold (the class straddles it, 80/100 stationary at $N = 12$). Bottom strip: the frozen null at machine precision. Ensemble verdicts are given in the text.

null-silence requirement (Sec. 2.3) demands that a witness vanish on the frozen class-(i) state. The out-of-time-order correlator fails: on the gapped ground state it reaches the arrival threshold at $t^* = 1.5$ and saturates at $C \approx 1.11$. Nothing is wrong with the OTOC as physics — it certifies operator spreading, which a frozen state supports — but that is precisely why it cannot certify that a *state* is moving. The preregistered discard clause executed, removing both OTOC statistics from the class-separation criterion.

The discard had teeth. With the OTOC gone, the original battery left exactly one class pair under-witnessed: scrambling vs. localized separated on Ξ alone (AUC 0.985/0.990 at the two sizes), with the Bohr participation ratio degrading with size ($0.89 \to 0.56$) and the registered recurrence statistic — the window *minimum* — landing at 0.71/0.945. Criterion (a), as originally registered, therefore **fails**. The counterfactual is recorded: had the OTOC been retained, its arrival time separates that pair at AUC 1.0 (localized fronts never arrive; scrambling fronts arrive at $t \approx 2$). The failure is single-cause, and it is a finding about witness design: the natural scrambling/localization discriminator is not null-silent, and the null-silent battery we registered was one statistic short.

The repair, and what fresh seeds taught us. The reformalized battery replaces the recurrence minimum — structurally uninformative for slow classes, since a slowly rotating state revisits its reference — with the recurrence *mean*, which is null-silent by construction and class-resolving in practice. On the data that motivated the redesign, all 18 pair-size checks passed; we labeled that table validation, not confirmation, and re-adjudicated on fresh committed seeds. At the registered ensemble size ($n = 20$) the fresh seeds **failed** a different pair — scrambling vs. integrable at $N = 10$, all three statistics between 0.85 and 0.93 — while passing at $N = 12$. The diagnosis is statistical, not physical: a sharp AUC threshold of 0.95 estimated from 20-vs-20 samples carries a standard error of 0.05–0.08, so near-threshold verdicts are seed-lotteries. Doubling the ensembles ($n = 40$, fresh seeds again) stabilized every margin: **all 18 pair-size checks pass**, the previously marginal pair at 0.952–0.979, and the repaired pair at 0.987–0.992 (the Bohr ratio for scrambling vs. localized remains honestly weak, 0.59 at $N = 12$ — two statistics carry the pair, as the criterion requires). A descriptive third size ($N = 14$, computed from eigenbasis coefficients alone) confirms the monotone trend: all three statistics ≥ 0.976.

The two methodological claims we take from this: null-silence is a nontrivial filter that the field's default dynamics diagnostic fails; and sharp-threshold criteria on small ensembles must be fresh-seeded and, near the threshold, power-stabilized — our original-seed validation table would have over-claimed.

5 Results III: class-resolved robustness and the dephasing sign structure

The robustness criterion asks whether metric stationarity responds to perturbation differently in different dynamical classes. Under local dephasing at the calibrated rate $\gamma = 0.01$, it does, with a sign structure that is the study's most striking single measurement (Fig. 4).

The mean log drift ratios, replicated at two sizes with disjoint confidence intervals and consistent direction: chaotic and scrambling +2.1 to +2.2 (noise strongly destabilizes their stationary metrics), localized +1.5, integrable +0.75 — and the coherence-carried classes *negative*: metastable −0.09/−0.11, quasiperiodic −0.27. Weak dephasing makes the quasiperiodic metric *more* stationary than its own unperturbed dy-

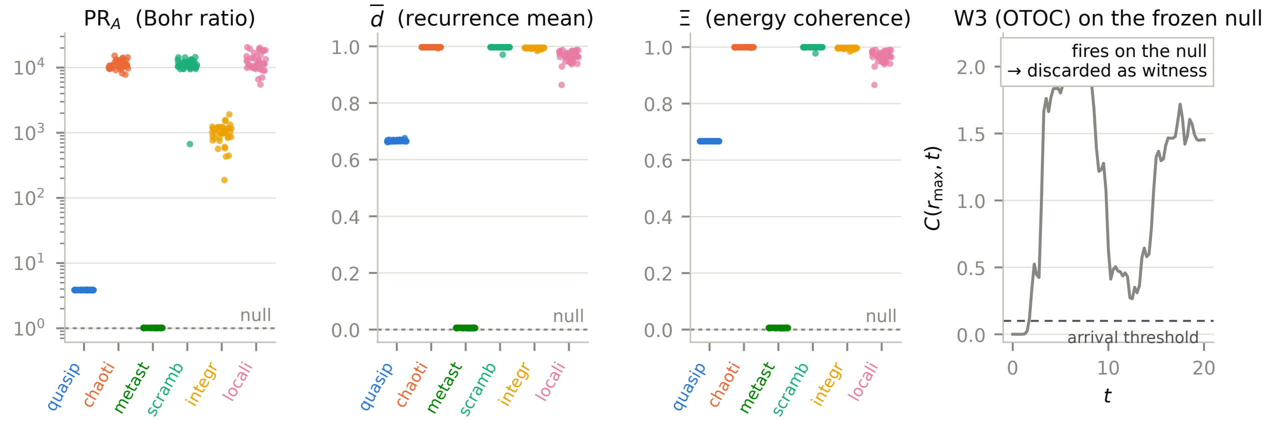

Figure 3: The final witness battery (fresh seeds, $n = 40$, $N = 12$; per-run values, one dot per run). Dashed lines: the frozen null's exact values. All three statistics are null-silent; class separation is visible directly. Right panel: the registered witness W3 (OTOC) evaluated on the frozen null — it crosses its own arrival threshold at $t^* = 1.5$ and saturates near 1.1–1.5, firing on a state that does not move; under the preregistered null-silence clause it was discarded from the criterion.

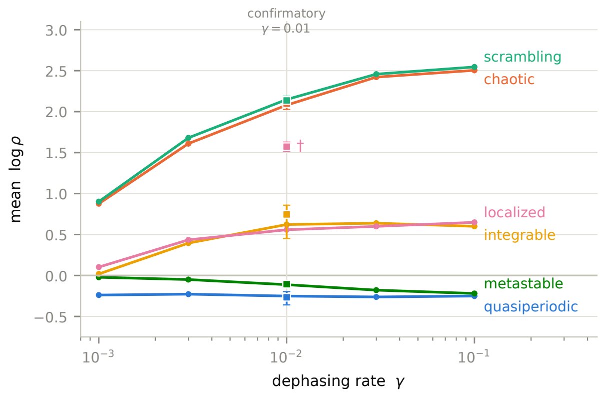

Figure 4: Dephasing response by class: mean $\log\rho$ versus rate γ (descriptive sweep, curves; first 10 manifest runs per class, Néel-primary for the localized regime). Squares with intervals: the confirmatory single-γ ensemble means with BCa 95% CIs. (†: the confirmatory localized ensemble uses 5 initial states per realization and reads higher than the Néel-primary sweep — an ensemble-composition difference, not a discrepancy.) The sign structure — negative for coherence-carried classes, strongly positive for chaotic ones — is stable over two decades and never reorders.

namics: the noise damps the coherences whose beating carries the metric's oscillation, pinning $D(t)$ to its time average. Every pairwise contrast required by the registered criterion clears the $\ln 1.5$ threshold in both tracks at both sizes; the class ordering (chaotic $\approx$ scrambling $>$ localized $>$ integrable $>$ metastable $\gtrsim$ quasiperiodic, the last two negative) is the pilot's ordering, confirmed at scale.

A descriptive strength sweep shows the sign structure is a regime, not a point: across two decades $\gamma \in [10^{-3}, 10^{-1}]$ the quasiperiodic response is flat at ≈ -0.25, the metastable response strengthens monotonically ($-0.02 \to -0.22$), the chaotic-class response rises and saturates ($+0.9 \to +2.5$), and the ordering never reorders. The mechanism is standard open-system physics — decoherence damping of coherence-carried oscillations, the weak-rate side of the Zeno family [6] — and we claim none of it as new. The class-resolved *geometric* reading — noise as a stabilizer of moving information metrics and a destabilizer of chaotic ones, with the sign tracking the dynamical class — is, to the extent of our survey [2, 3, 7, 8], unreported; we state this conditionally and note the survey was not systematic.

The other two protocols are clean nulls at the registered strengths: the local quench leaves every dynamical class's drift ratio at $|\overline{\log\rho}| \leq 0.07$, and two-site loss reads 0.07–0.25 everywhere. (The fixed point's large quench ratio, $+4.4$, is floor-referenced — its unperturbed drift is exactly zero — and carries no class information; instrument note in Appendix B.) All robustness statements are statements at the calibrated strengths and, per the sweep, within the probed regime — not strength-independent class properties.

6 Results IV: the motionless-comparator search — one class survives

The sharpest form of the sustained-by question is: *does there exist a motionless state carrying the same time-averaged metric?* If yes, the dynamics is optional for the structure; if no, the structure is motion-borne. This section reports the search, including the two corrections our own audits forced (Fig. 5).

The canonical comparator is not matched. The diagonal ensemble $\bar{\rho}$ — the motionless state with the run's exact time-averaged two-site reduced density matrices — was registered as metric-matched by construction. It is not: mutual information is nonlinear in the reduced density matrices, and $\Phi[\bar{\rho}]$ sits 43–90% of $\|\bar{D}\|$ away from the run mean, depending on class. Relatedly, our first robustness comparison against $\bar{\rho}$ produced a large "the comparator is more fragile" differential that an adversarial re-analysis exposed as a floored-denominator artifact (the comparator has no baseline drift to compare against; its absolute perturbed response is in fact *smaller*). Both corrections are dated in the study record; what survives them is stronger than what they replaced.

The switch-off measurement. Dephasing the state mid-window — which projects exactly onto $\bar{\rho}$ — moves the metric by 43–90% of $\|\bar{D}\|$ (chaotic 0.54, scrambling 0.52, integrable 0.85, metastable 0.90, localized 0.90, quasiperiodic 0.43). Killing the motion does not freeze the metric in place; it changes the metric. For the metric-stationary classes this registers as a drift increase with disjoint confidence intervals.

The optimization search, hardened. We then searched explicitly for *any* motionless match: stationarity means $[\sigma, H] = 0$, so we optimized over populations on the energy eigenbasis — three natural families (thermal at both temperature signs, depolarized diagonal ensemble, Gaussian microcanonical) and a general smooth-$f(H)$ family (log-populations parameterized by up to 24 Chebyshev coefficients, multi-start optimization), over full ensembles at both criterion sizes. A first, narrower version of this search (families only, one run per class) wrongly suggested the metastable and integrable classes were unmatchable; the hardened search corrected this before drafting, and the correction is part of the record.

The final result is a single-survivor split. Matchable within ε_Φ: chaotic and scrambling in every run, almost exactly (median miss 0.035–0.046 — a Gaussian microcanonical window reproduces the time-averaged metric of a chaotic state, as eigenstate thermalization would suggest); metastable in every run (0.09–0.13); integrable in every run (0.16–0.17); localized as a boundary case (3–6 of 20 runs, median ≈ 0.27). Unmatched: **the quasiperiodic class, in 0 of 80 runs across both sizes** (median miss 0.31; the single closest run reaches 0.264, still above threshold), robustly across mode numbers $m = 3, 4, 5$ (medians 0.31–0.33) and insensitive to enriching the optimizer. The claim is bounded by its search space — stationary states that are smooth functions of H — and is not an impossibility theorem; within that space, it is unambiguous.

Two independent instruments therefore single out the same class. The quasiperiodic metric is the only one that no searched motionless state can carry, and the only one that weak noise *stabilizes*. Incommensurate coherence-carried motion is, in this family, the regime whose emergent structure needs its dynamics: motion-borne in the operational sense this paper defines. Conversely, the chaotic classes' stationary metrics are compatible-with their motion — a thermal-window state carries the same structure — while their *robustness signature* still distinguishes them (Sec. 5): the two probes answer different questions, and both answers are class-resolved.

7 Results V: the driven regime

The discrete-time-crystal track supplies the study's cleanest stationary-with-witness regime and its most instructive compatible-with verdict (Fig. 6).

At $\varepsilon = 0.03$, deep in the rigid phase, all 100 runs are metric-stationary while the subharmonic witness reads 0.939: a period-doubled magnetization response, locked across 20 disorder realizations, above a mutual-information metric that does not move. Stationarity degrades with drive imperfection (88/100 at $\varepsilon = 0.06$, 52/100 at 0.10), tracking the witness. The rigidity curve is measured over its full range: the subharmonic weight holds above 0.65 through $\varepsilon = 0.20$, crosses 1/2 at $\varepsilon_c \approx 0.23$ at our drive parameters, and collapses by $\varepsilon \approx 0.45$ — a complete critical-strength measurement for this realization of the protocol

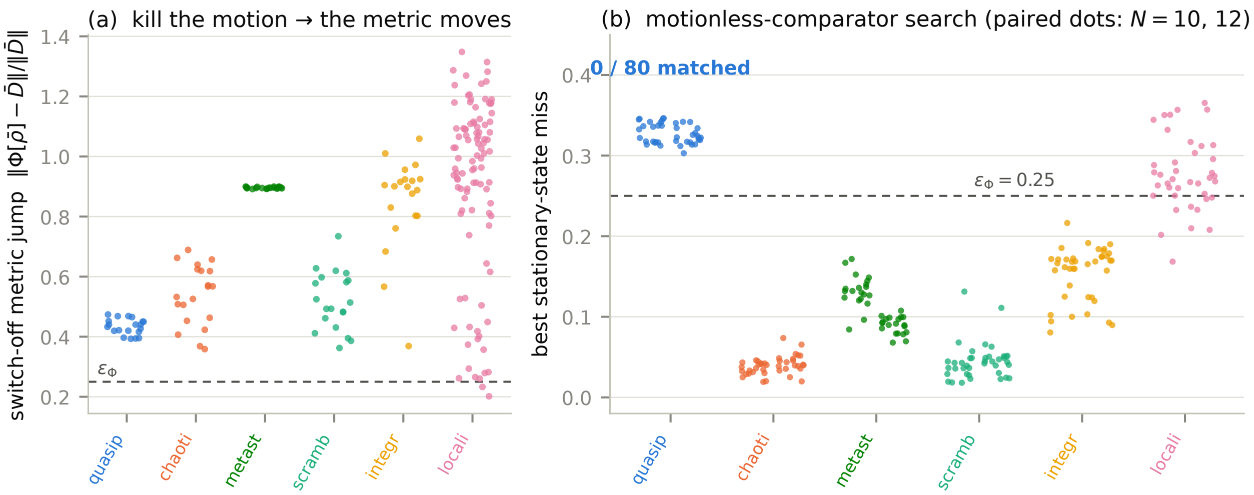

Figure 5: (a) The switch-off measurement: dephasing the state mid-window (equivalently, projecting onto the diagonal ensemble) moves the metric by 43–90% of $\|\bar{D}\|$ in every class — killing the motion changes the metric. (b) The hardened motionless-comparator search: best achievable miss $\|\Phi[\sigma] - \bar{D}\|/\|\bar{D}\|$ over three natural stationary families plus a general smooth-$f(H)$ optimization, every run, paired dots per class ($N = 10$ left, $N = 12$ right). Every class is matchable within ε_Φ except the quasiperiodic class (0/80 including the stress test); localized is the boundary case.

of Refs. [4, 5].

The switch-off test resolves the sustained-by question for this regime, and resolves it *against* the drive: removing the kicks at mid-window collapses the witness ($0.95 \to 0.00$) while the metric's stationarity persists and slightly improves (mean post-window drift $0.110 \to 0.086$). The frozen-in disorder, not the drive, holds this metric — the witnessed subharmonic motion is compatible with the stationary structure, not required by it. We registered this as an open question with exactly this outcome listed as reportable, and report it accordingly. Consistently, the perturbation protocols at the calibrated strengths barely move the DTC metric ($\overline{\log \rho} = 0.01$–$0.19$).

The comparator regimes behave asymmetrically, and honestly. Without interactions (r1), the subharmonic peak is destroyed at $\varepsilon = 0.03$ (0.13) — the fine-tuned control fails as expected, certifying that interactions produce the rigidity. Without disorder (r2), however, the clean interacting drive does *not* thermalize on any horizon we probed: its subharmonic response persists undiminished (0.89–0.90) to 2000 periods. At small ε the r2 regime is not a usable thermalizing control, and the discriminating comparator burden falls entirely on r1 — a comparator-scope finding we report rather than bury.

8 Scope walls and limitations

The metric belongs to the factorization. Everything here is computed on the natural site partition, and the dependence on that choice is not small and not uniform: under fixed nonlocal two-site frame changes, the metric moves by a class-dependent amount — mean relative change 0.11 (chaotic), 0.15 (scrambling), 0.20 (localized), 0.22 (integrable), 0.42 (quasiperiodic), 0.53 (metastable, maximum observed 0.82). The coherence-carried classes — including the survivor class of Sec. 6 — have the most frame-dependent metrics. This does not undermine the within-frame comparisons (every matchability and robustness statement compares objects in the same partition, which transforms both sides together), but it bounds the interpretation: these are properties of the metric-on-a-factorization, not of the state alone. Notably, strictly local (single-site) frame changes leave the metric invariant to machine precision — the dependence lives entirely at the partition level, which is where the construction's own authors located their caveat [1].

Finite size, finite family, finite search. Criterion verdicts are established at $N \in \{10, 12\}$ with a descriptive $N = 14$ point; the quasiperiodic construction is exhaustively unsatisfiable at $N = 8$ (0/56 candidate mode triples), which is a

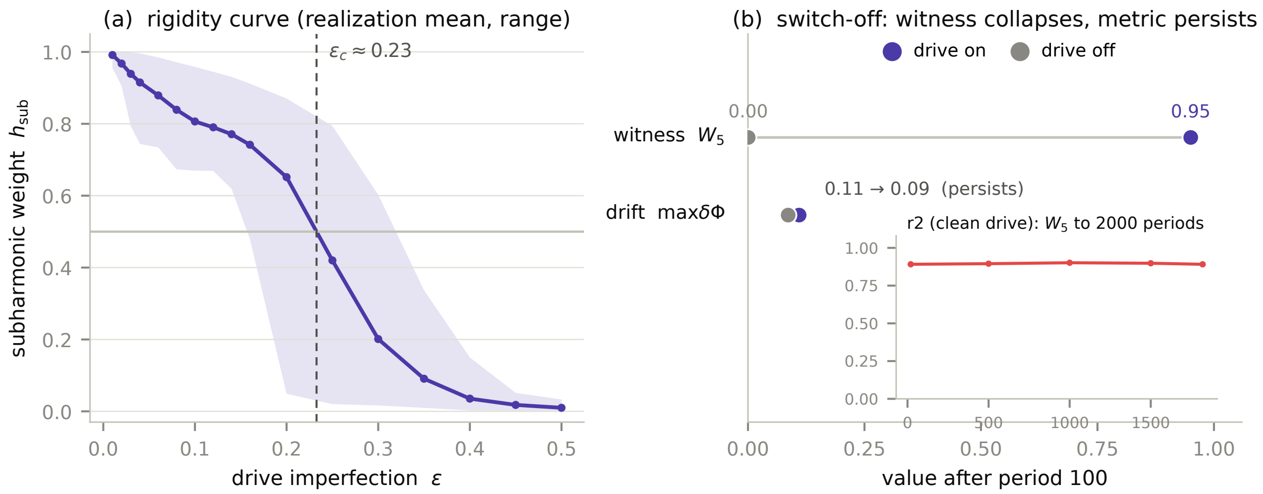

Figure 6: (a) The full rigidity curve of the driven track: subharmonic weight $h_{\rm sub}$ versus drive imperfection (realization mean and range, seed-paired across ε), crossing $1/2$ at $\varepsilon_c \approx 0.23$ (interpolated). (b) The switch-off test at $\varepsilon = 0.03$: removing the drive collapses the subharmonic witness while the metric drift persists essentially unchanged — localization, not the drive, holds this metric. Inset: the clean interacting comparator (r2) does not thermalize on any probed horizon; its subharmonic weight is undiminished at 2000 periods.

warning about small-size instantiations of incommensurability generally. The stationarity threshold $\varepsilon_\Phi = 0.25$ is calibrated to this construction's finite-size noise floor and has no significance beyond it. The comparator search spans stationary states that are smooth functions of H; fine-tuned non-smooth populations are outside it, and the unmatchability result is bounded accordingly. The localized regime straddles the stationarity threshold (80/100) and we have kept it out of every headline count. Robustness statements hold at the calibrated strengths and within the swept regime ($\gamma \in [10^{-3}, 10^{-1}]$). All clocks are external laboratory clocks; nothing here addresses systems that must supply their own.

9 Discussion and outlook

What the study delivers is not a mechanism but an instrument: the instantiated-vs-maintained distinction, usually a rhetorical flourish, is here an executable measurement with three independent probes — and its first execution returned a sharp, doubly-corroborated answer. In this family, thermalizing motion builds structures a motionless thermal state could carry; incommensurate coherent motion builds one that nothing motionless we searched can carry, and that noise defends rather than destroys.

The result points directly at a next question: is there a form of robustness that *recurrent* or quasiperiodic dynamics provides and fixed points cannot? The two signatures reported here — unmatchability and noise-stabilization, coinciding in one class — are exactly the evidence such a mechanism would leave, and the driven track supplies a controlled arena in which to look for it (with the caution of Sec. 7: the most spectacular witnessed motion in this study turned out *not* to sustain its metric).

The test template itself is not specific to spin chains. Any system with a computable structural functional over a dynamical substrate admits the same three probes: certify the motion with null-silent witnesses, switch the dynamics off, and search honestly for a motionless impostor. Candidate applications suggest themselves wherever stable emergent structure rides on flux — resting-state functional connectivity over neural dynamics, metabolic network structure over flux-carrying steady states, stability of ecological or market structure over turnover — and we offer the template for these settings as a proposal only: nothing beyond spin chains has been demonstrated here. The distinction the test operationalizes — structure that merely exists versus structure that is actively kept in existence — is, of course, much older than physics; we make no attempt to engage that literature here.

On the methods side, we draw two conclusions we believe generalize. Null tests for witnesses are cheap and merciless — the OTOC episode suggests that diagnostics certifying "dynamics" deserve routine auditing against frozen states. And preregistration with committed seeds changed the outcome of this study at least four times; the amendment log (Appendix C) is our answer to the question of what such discipline buys in a purely computational setting.

The obvious open front is scale and covariance: whether any version of this question can be asked of structures that deserve the word "geometry" with fewer scare quotes — partition-covariant functionals, larger systems, internally clocked dynamics — remains exactly as open as it was, and none of the present results shortens that road. What they do establish is that the road's first step is measurable: there exist, already in twelve qubits, emergent structures that demonstrably need their dynamics.

Acknowledgements. The numerical study — implementation, execution, analysis, and the adversarial companion — was carried out with Claude (Anthropic) as AI research assistant, operating under the preregistration and shared-substrate protocol whose record forms Appendix C. The programme's initiating knowledge base was drafted in collaboration with OpenAI's ChatGPT. All rulings, decisions, and the final text are the author's responsibility.

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Data and code availability. The full study record — specification, seed manifests, amendment log, evidence packets, session logs, analysis code, and all run outputs — is available in the public repository https://github.com/olegroshka/ideg, pinned at the submission tag (`paper-v1`).

A Numerics

Exact diagonalization throughout (dense to $N = 14$, dimension 16384); pure-state evolution via eigendecomposition, density-matrix evolution for the dephasing protocol via Trotter splitting of the unitary and the exact per-step damping mask (step 0.5), $N \le 10$. State-norm drift across all unitary runs $< 4 \times 10^{-15}$; trace drift of dephased runs recorded per run at the same scale. Python 3.11 / NumPy 2.3 / SciPy 1.15; every random draw seeded from committed manifests; total compute across pilot, confirmatory, adversarial, and hardening campaigns of order a machine-day on a 24-core workstation. The $N = 14$ witness point uses the identity $|\langle\psi(t_0)|\psi(t)\rangle|^2 = |\sum_n p_n e^{-iE_n(t-t_0)}|^2$, so recurrence statistics require only eigenbasis coefficients. The comparator search precomputes per-eigenstate two-site reduced density matrices, making each candidate stationary state's metric a population-weighted sum evaluable in milliseconds at any dimension.

B Statistics

Class separation: exact Mann–Whitney AUC, symmetrized $\max(A, 1-A)$ since direction is not registered; threshold 0.95; ≥ 2 statistics per pair; two sizes; singleton classes (the fixed point) by exact-value exclusion (the ensemble range must exclude the deterministic value). Robustness: mean log drift ratio per class, BCa bootstrap (1000 resamples, 95%), disjoint intervals plus $|\Delta| > \ln 1.5$, replicated with consistent direction; disordered ensembles resampled at the realization level. Seed sensitivity: at $n = 20$-vs-20 the AUC standard error near $A \approx 0.9$ is 0.05–0.08, so verdicts within one standard error of threshold are lotteries — measured directly here when fresh seeds flipped a marginal pair (Sec. 4); $n = 40$ halves the standard error and stabilized every margin. Size trend for the marginal pair (AUC at $N = 10 \to 12 \to 14$): Bohr ratio $0.979 \to 0.979 \to 1.000$; recurrence mean $0.952 \to 0.975 \to 0.976$; energy coherence $0.956 \to 0.975 \to 0.976$. Instrument notes: the log-ratio floor (10^{-3}) makes exactly-stationary objects read $\log\rho \approx \log(\delta\Phi_{\mathrm{pert}}/10^{-3})$ — class-(i) quench values are floor-referenced and excluded from class inference; the metric's $-\ln x$ weight cap amplifies machine-epsilon mutual-information jitter by up to $x_{\min}^{-1} = 10^6$, setting an irreducible $\sim 10^{-9}$ numerical floor on metric-space identity checks for near-product states.

C The amendment log

The complete dated record of every post-registration change, each with its reason and repository commit; no entry alters data already taken.

D Invariance battery

Every witness statistic and the metric were checked under: global phase; consistent local basis change (state, Hamiltonian, and operators together); local basis change of the state alone; and site reflection. Witness statistics and mutual-information matrices are identical within 10^{-10} in all mandatory items across every class and size (the participation ratio judged relative to its own magnitude, which reaches 10^5–10^6 at $N = 12$). Metric-space deviations carry the cap-amplified numerical floor described in Appendix B and are reported, not thresholded. The state-only local change leaves the metric invariant to machine precision — single-site entropies are local-frame invariant — which localizes all factorization dependence at the partition level; the partition probe of Sec. 8 (fixed random entangling two-site frames on three disjoint pairs, 24–72 samples per class) is the quantitative version.

E Driven-track implementation notes

Disorder realizations and initial states are seed-paired across drive strengths and comparator regimes (identical couplings and fields at equal seed), so rigidity curves and regime contrasts are paired comparisons. The subharmonic weight is computed on an even number of stroboscopic periods so the period-doubled line is the exact Nyquist bin. Switch-off evolves under the interaction Hamiltonian alone in lab time, sampled at the same period boundaries (drive removed, clock kept). The dephasing protocol applies the exact per-period damping mask with the rate stated in lab-time units ($T_{\text{period}} = 2$).

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date	entry	commit
08-11	Spec registered; threshold review (Am. 1): criterion-(b) instrument $\to$ log drift ratio + calibration pilot	`24eaa858ec60` `7e40011d56c8`
08-11	Am. 2 (pre-run): integrable certificate Poisson $\to$ free-spectrum reconstruction (mis-specification measured: free chain)	`602658b1b96c`
08-11	Am. 3 (pilot, owner-ruled): ε_Φ 0.05 $\to$ 0.25 (below measured noise floor); confirmatory $\lambda = 0.1$, $\gamma = 0.01$	`5db69bade460`
08-12	Instrument survey: KEEP log-ratio (no change; dated entry)	`27ada59e697c`
08-12	Confirmatory manifest committed pre-execution; survey window closed at first run	`b91ad54da91f`
08-12	Addendum 1: quasiperiodic construction unsatisfiable at $N = 8$ (0/56, exhaustive) $\to$ criterion sizes (10, 12)	`a76db0e67e09`
08-12/13	Confirmatory executed: W3 fires on null $\to$ discarded; (a) fails as registered; (b) holds; verdicts of record	`fbe324f38f34`
08-13	Adversarial companion: fragility-direction retraction (floored denominator); comparator matching assumption refuted	`33fdd6287776`
08-13	Am. 4 (ratified): battery $\to \{\mathrm{PR}_A, \overline{d}, \Xi\}$; W3 descriptive; fresh-seed re-adjudication required	`9030c0ebdab6`
08-13	Fresh seeds $n = 20$: FAIL on a different, threshold-marginal pair (seed lottery measured)	`5efaa7309c56`
08-13	Am. 5 (ratified): $n = 40$, final verdict accepted either way $\to$ (a) HOLDS 18/18	`3e7c0a787a09`
08-13	Hardened comparator search corrects family-only over-claim: single-survivor result	`67dda68970b1`
08-13	Descriptive hardening sweep: ε_c, 2000-period comparator, γ-grid, partition distribution, $N = 14$	`9ce17dec803c`

Table 2: The amendment log (12-character commit identifiers in the public repository).